Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
5	(a)			Fusion definition (nuclei join to form larger nucleus and energy release) (1) Fission definition (larger nucleus splits to smaller nuclei and energy release) (1) N.B. energy release only required somewhere once Products are (usually) more stable / have higher BE/N (1) Low A numbers do fusion, high A numbers do fission (might be in diagram with arrows) (1)	4			4		
	(b)	(i)		The Avogadro constant is the {number of / 6.02×10^{23}} {particles / atoms / molecules} in 1 mole of a substance **or** in 12 g of ${}^{12}_{6}\mathrm{C}$	1			1		
		(ii)		LHS - RHS i.e. (235.04393 + 1.00866) – (97.91273 + 134.91645 + 3 $\times$1.00866) = 236.05259 – 235.85516 = 0.19743 [u] (1) $\times$ 931 or 1.66×10^{-27} (1) Convert from eV into J **or** kg into J i.e. 2.94×10^{-11} [J] (1) Method for number of atoms e.g. $\frac{(6.02\times10^{23})}{235}\,1 = 2.56 \times 10^{21}$ **or** $\frac{0.001}{235 \times 1.66 \times 10^{-27}}$ (1) Energy released = 7.5×10^{10} [J] (1)		5		5	5	
				Question 5 total	**5**	**5**	**0**	**10**	**5**	**0**